SIM65M and SIM68M_Compatible_Design

GNSS Module

SIMCom Wireless Solutions Limited
Building B, SIM Technology Building, No.633, Jinzhong Road
Changning District, Shanghai P.R.China
Tel: 86-21-31575100
support@simcom.com
www.simcom.com

Document Title:	SIM65M and SIM68M_Compatible_Design
Version:	V1.00
Date:	2022-8-15
Status:	Released

GENERAL NOTES

SIMCOM OFFERS THIS INFORMATION AS A SERVICE TO ITS CUSTOMERS, TO SUPPORT APPLICATION AND ENGINEERING EFFORTS THAT USE THE PRODUCTS DESIGNED BY SIMCOM. THE INFORMATION PROVIDED IS BASED UPON REQUIREMENTS SPECIFICALLY PROVIDED TO SIMCOM BY THE CUSTOMERS. SIMCOM HAS NOT UNDERTAKEN ANY INDEPENDENT SEARCH FOR ADDITIONAL RELEVANT INFORMATION, INCLUDING ANY INFORMATION THAT MAY BE IN THE CUSTOMER'S POSSESSION. FURTHERMORE, SYSTEM VALIDATION OF THIS PRODUCT DESIGNED BY SIMCOM WITHIN A LARGER ELECTRONIC SYSTEM REMAINS THE RESPONSIBILITY OF THE CUSTOMER OR THE CUSTOMER'S SYSTEM INTEGRATOR. ALL SPECIFICATIONS SUPPLIED HEREIN ARE SUBJECT TO CHANGE.

COPYRIGHT

SIMCom Wireless Solutions Limited
Building B, SIM Technology Building, No.633 Jinzhong Road, Changning District, Shanghai P.R.China
Tel: +86 21 31575100
Email: simcom@simcom.com

For more information, please visit:
https://www.simcom.com/download/list-863-en.html

For technical support, or to report documentation errors, please visit:
https://www.simcom.com/ask/or email to: support@simcom.com

Version History

Date	Version	Description of change	Author
2022-08-15	1.00	Original	Xinao Liu Wentao Tan

Contents

Table Index

Figure Index

1. Introduction

This document is targeted for customers to understand the differences between SIM65M and SIM68M. Customers can use SIM65M or SIM68M module to design and develop applications quickly.

2. PIN assignment

2.1 Pin Assignment Overview

The following figure shows the pin assignment of SIM65M and SIM68M.

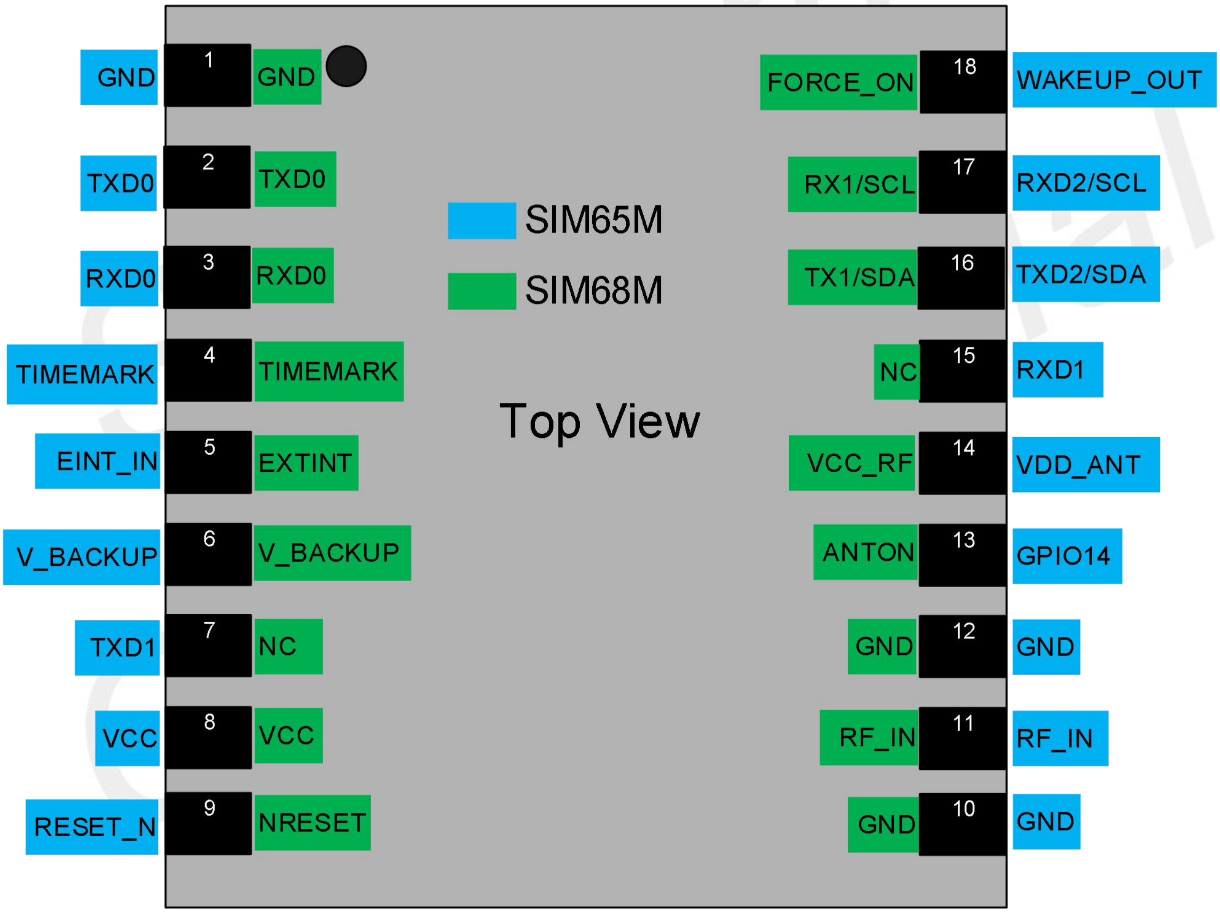

Figure 1: SIM65M and SIM68M pin assignment (Top view)

2.2 Differences Overview

Table 1: The differences overview

Functions	SIM65M	SIM68M

UART0	Support NMEA and update	Support NMEA and update
UART1	Support System Log	Support RTCM function
UART2	Support RTCM function	Not support
PPS	Support	Support
RESET	Support	Support
Sleep mode	Support	Support
RTC mode	Support	Support
Antenna detect	Not support	Not support
IO power domain	2.8V	2.8V

2.3 Differences of Electronic Characteristic

Table 2: The differences of electronic characteristic

Pin #	SIM65M		SIM68M	
	PIN name	Voltage range	PIN name	Voltage range
8	VCC	2.8~4.3V	VBAT	2.8~4.3V
6	V_BACKUP	2.5~3.6V	V_BACKUP	2.0~4.3V
2	TXD0	2.8V	TXD0	2.8V
3	RXD0	2.8V	RXD0	2.8V
16	TXD2/SDA	2.8V	TXD1/SDA	2.8V
17	RXD2/SCL	2.8V	RXD1/SCL	2.8V
4	TIMEMARK	2.8V	TIMEMARK	2.8V
5	EINT_IN	2.8V	EXTINT	2.8V
9	RESET_N	2.8V	NRESET	2.8V
7	TXD1	2.8V	NC	/
13	GPIO14	2.8V	ANTON	2.8V
14	VDD_ANT	2.8V	VCC_RF	2.8V
15	RXD1	2.8V	NC	/
18	WAKEUP_OUT	2.8V	FORCE_ON	2.8V

NOTE

1. For details information, please refer to each HD guide

Table 3: Difference in Pin Definitions

Pin #	SIM65M	SIM68M
5	EINT_IN	EXTINT
7	TXD1	NC
8	VCC	VBAT
13	GPIO14	ANTON
15	RXD1	NC
18	WAKEUP_OUT	FORCE_ON

3. Recommended Footprint

3.1 Top and Bottom View

The following figures show top and bottom view of SIM65Mand SIM68M.

There has no difference for footprint.

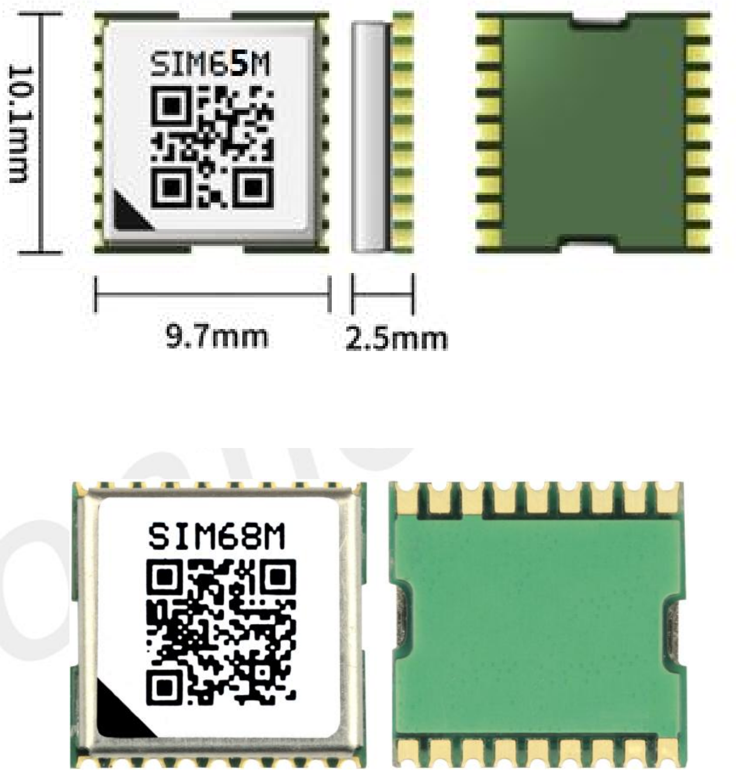

Figure 2: SIM65M and SIM68M top and bottom view

3.2 Recommended Stencil Design

SIM65M and SIM68M have the same recommended stencil design.

The recommended stencil design for SIM65M and SIM68M is shown as below.

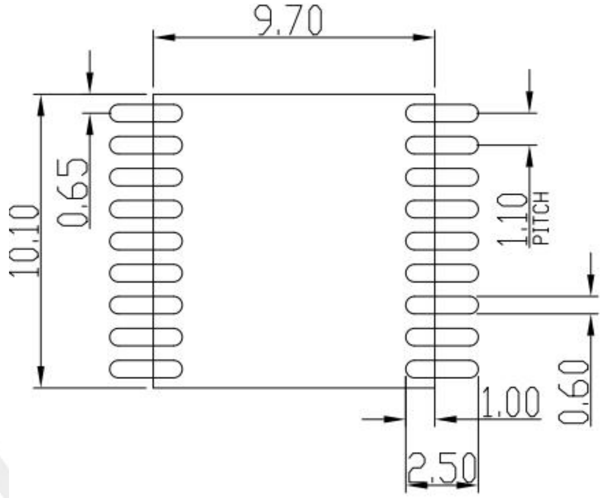

Figure 3: Recommended Stencil Design for SIM65M and SIM68M (Unit: mm)

4. Hardware Reference Design

This chapter introduces compatible design between SIM65M and SIM68M on main functionalities.

4.1 Power Supply

VCC power supply

The power supply pins of SIM65M and SIM68M have one VCC pin (pin 8) ,VCC pin supply the main power for module. The following figure is the reference design of the module VCC power supply.

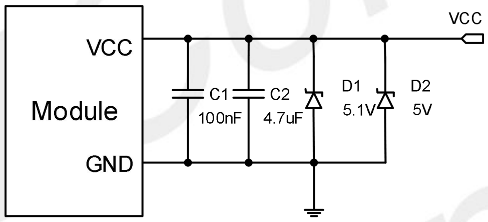

Figure 4: Power supply reference circuit

Table 4: The VCC power range

Module	VCC power supply			VCC power peak current
	Min	Typical	Max	Max
SIM65M	2.8V	3.3V	4.3V	100mA
SIM68M	2.8V	3.3V	4.3V	100mA

Power design for a module is critical to its performance. The power supply of SIM65M and SIM68M should be able to provide sufficient current up to 100mA.

NOTE

For details information, please refer to each HD guide

V_BACKUP power supply

The V_BACKUP power supply range of SIM65M and SIM68M is shown in the table below, suggest customer keep the V_BACKUP supply active all the time, the module will perform a quick start every time it is power-on.

Table 5: The V_BACKUP power range

Module	V_BACKUP power supply			V_BACKUP power peak current
	Min	Typical	Max	Max
SIM65M	2.5V	2.8V	3.6V	10mA
SIM68M	2.0V	3.0V	4.3V	10mA

4.2 NRESET

The NRESET pin (active low) is used to reset the system, normally external control of NRESET is not necessary. The signal can be left floating, if not used. When NRESET signal is used, it will force volatile RAM data loss. Note that Non-Volatile backup RAM content is not cleared and thus fast TTFF is possible. The input has internal pull up.

4.3 PPS OUT

The PPS pin outputs pulse-per-second (PPS) pulse signal for precise timing purposes after the position has been fixed. The PPS signal can be provided through designated output pin for many external applications. This pulse is not only limited to be active every second but also allowed to set the required duration, frequency, and active high/low by programming user-defined settings.
PPS GPS time reference with adjustable duty cycle , support for time service application, which is achieved by the PPS vs NMEA feature.

The following figure is the typical application of the PPS function.

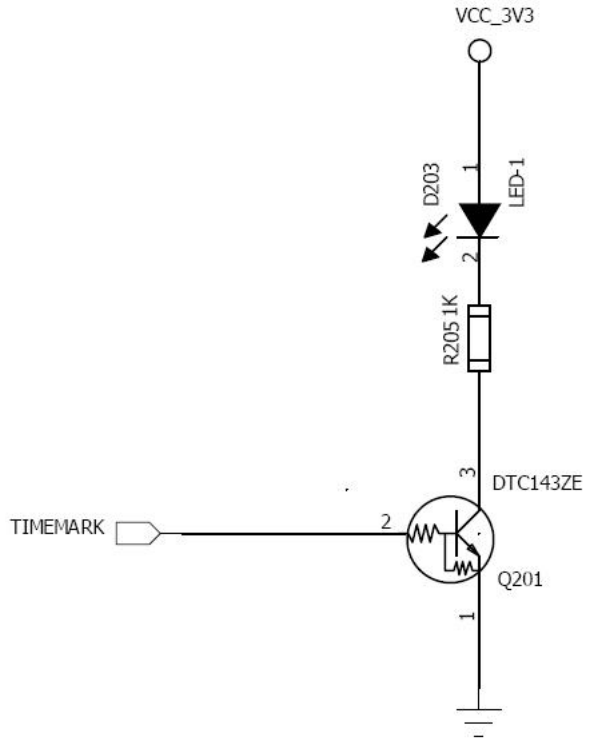

Figure 5: TIMEMARK application circuit

4.4 UART Interface

The module is as the DCE (Data Communication Equipment) and the client PC is as the DTE (Data Terminal Equipment).

SIM68M includes two UART (UART0 and UART1) interface for serial communication, and the SIM65M includes three UART(UART0, UART1 and UART2) interface for serial communication. The UART0 is as NMEA output and command input. SIM65M UART1 is used to capture system logs. SIM68M UART1 is as RTCM output. SIM65M UART2 is as RTCM output.

Below are the reference circuits.

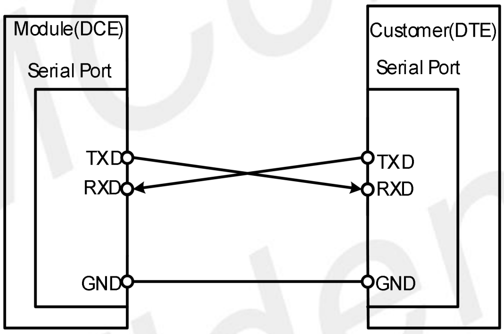

Figure 6: UART Full modem

Table 6: UART interface description

Module	UART	Votage	Description	Comment
SIM65M	UART0	2.8V	NMEA output, command input, firmware upgrade	2.8V power domain
	UART1	2.8V	System LOG output/ input	2.8V power domain, If unused, keep open.
	UART2	2.8V	RTCM output	2.8V power domain, If unused, keep open. Can be configured as I2C by software
SIM68M	UART0	2.8V	NMEA output, command input, firmware upgrade	2.8V power domain
	UART1	2.8V	RTCM output	2.8V power domain, If unused, keep open. Can be configured as I2C by software

4.5 Antenna Interface

SIM65M and SIM68M provide an antenna interface.

The antenna is a critical item for successful GPS reception in a weak signal environment. Proper choice of the antenna will ensure that satellites at all elevations can be seen, and therefore, accurate fix measurements are obtained.

It is recommended to use an active GPS antenna. In a typical application, SIM65M or SIM68M with an active antenna can get a tracking sensitivity about 3dB better than SIM65M with a passive antenna.

It is suggested the active antenna should be chosen as following:

Table 7: Antenna Specifications

Parameter	Specification	
Passive Antenna Recommendations	Frequency range	1575±3MHz
	Polarization	RHCP & Linear
	Gain	> 0dBi
	VSWR	< 2
Active Antenna Recommendations	Frequency range	1575±3MHz
	Polarization	RHCP & Linear
	VSWR	< 2
	Noise Figure	< 1.5dB
	Gain	> -2dBi

4.5.1 Antenna Choice and RF Design Consideration

To obtain excellent GPS reception performance, a good antenna will always be required. The RF circuits should also be designed properly based on the type of antenna.

The antenna design part of SIM65M and SIM68M is the same,The following is an example of SIM65M to illustrate the relevant design.

Passive Antenna

Passive antenna contains only the radiating element, e.g. the ceramic patch, the helix structure, and chip antennas. Sometimes it also contains a passive matching network to match the electrical connection to 50

Ohms impedance.

The most common antenna type for GPS/GLONASS/BEIDOU application is the patch antenna. Patch antennas are flat, generally have a ceramic and metal body and are mounted on a metal base plate.
Figure 7 shows a minimal setup for a GPS/GLONASS/BEIDOU receiver with SIM65M module.

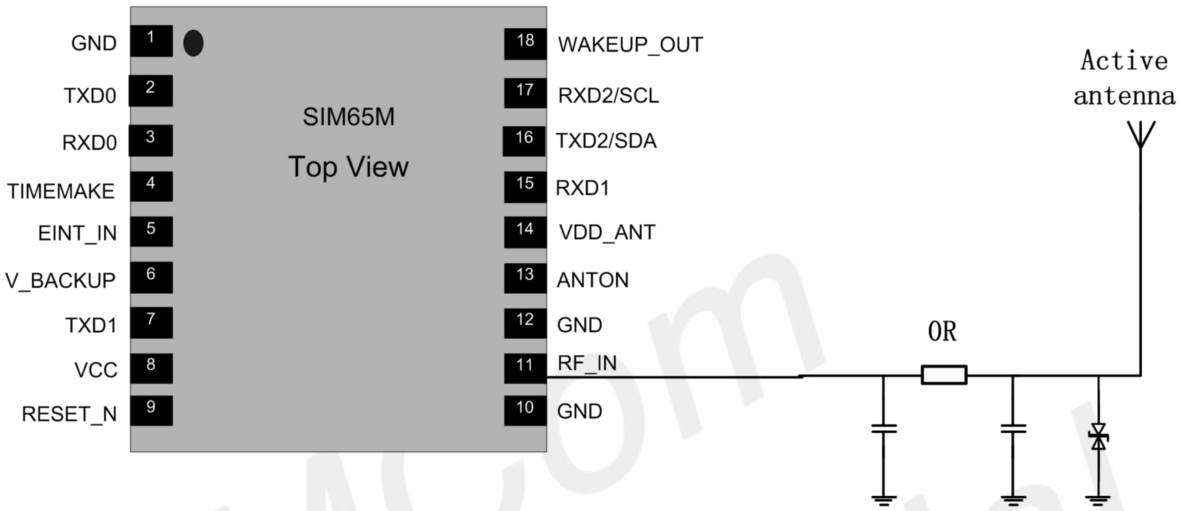

Figure 7: SIM65M passive antenna design

For better performance with passive antenna designs user can use an external LNA to increase the sensitivity up 3~4 dB. Please see Figure 8.

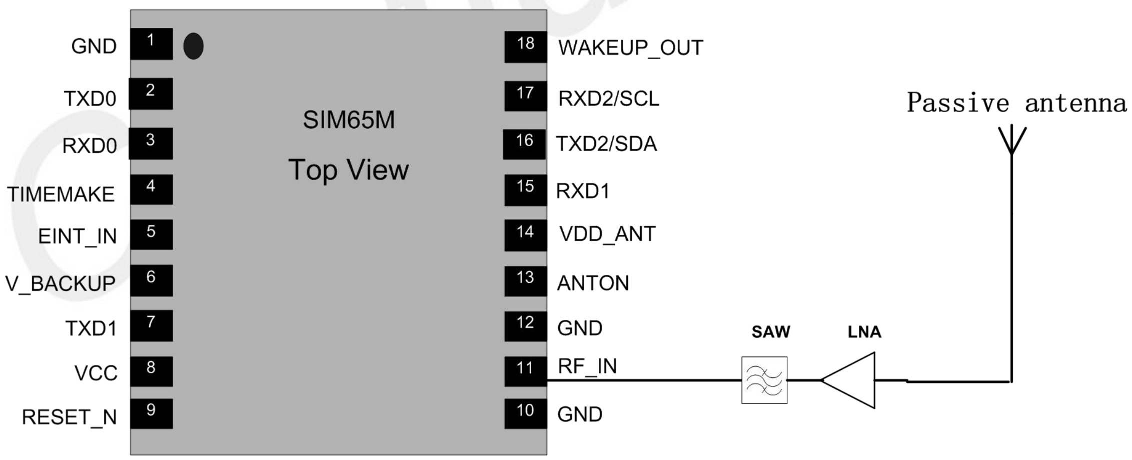

Figure 8: SIM65M passive antenna design (with external LNA and SAW)

Active Antennas

Active antenna has an integrated Low-Noise Amplifier (LNA). Active antenna needs a power supply that will contribute to GNSS system power consumption.

Usually, the supply voltage is fed to the antenna through the coaxial RF cable shown as Figure 9. The output voltage of PIN 14 is 2.8V. If the supply voltage of active antenna is 2.8V, PIN 14 VDD_ANT can be connected to RF_IN as figure 14 shows. If the active antenna is not 2.8V, other power should be connected to RF_IN.

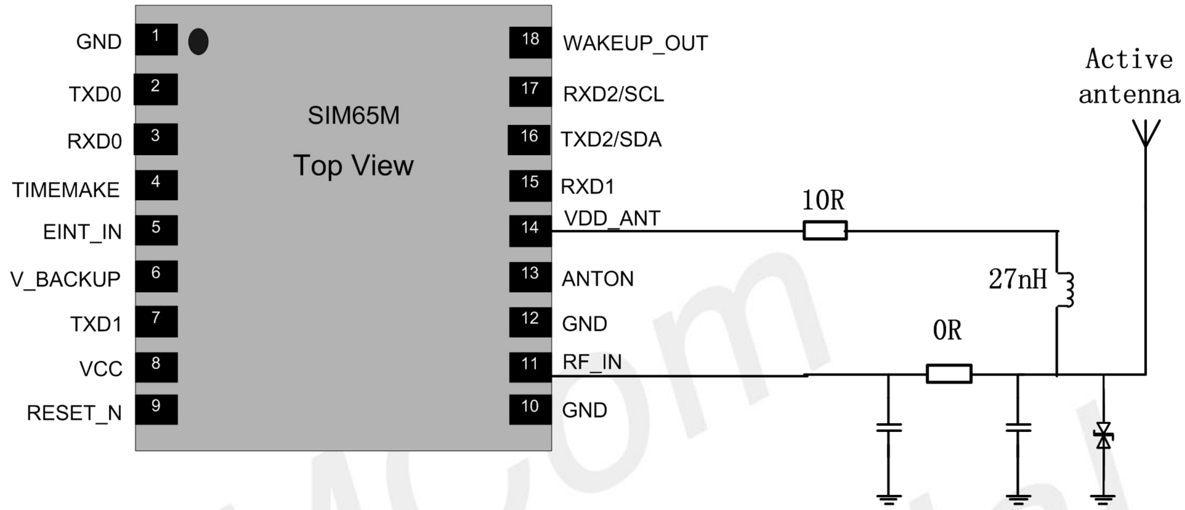

Figure 9: SIM65M Active antenna simplified design

If the customer's design is for automotive applications, then an active antenna can be used and located on top of the car in order to guarantee the best signal quality.

GNSS antenna choice should base on the designing product and other conditions. For detailed Antenna designing consideration, please refer to related antenna vendor's design recommendation. The antenna vendor will offer further technical support and tune their antenna characteristic to achieve successful GNSS reception performance depending on the customer's design.

4.6 Antenna Detection Design

Following figure is the typical application of SIM65M with active antenna which supplied by VCC_RF. If customer applies other kind of active antenna, keep PIN 14 floating and connect other voltage to the R125.

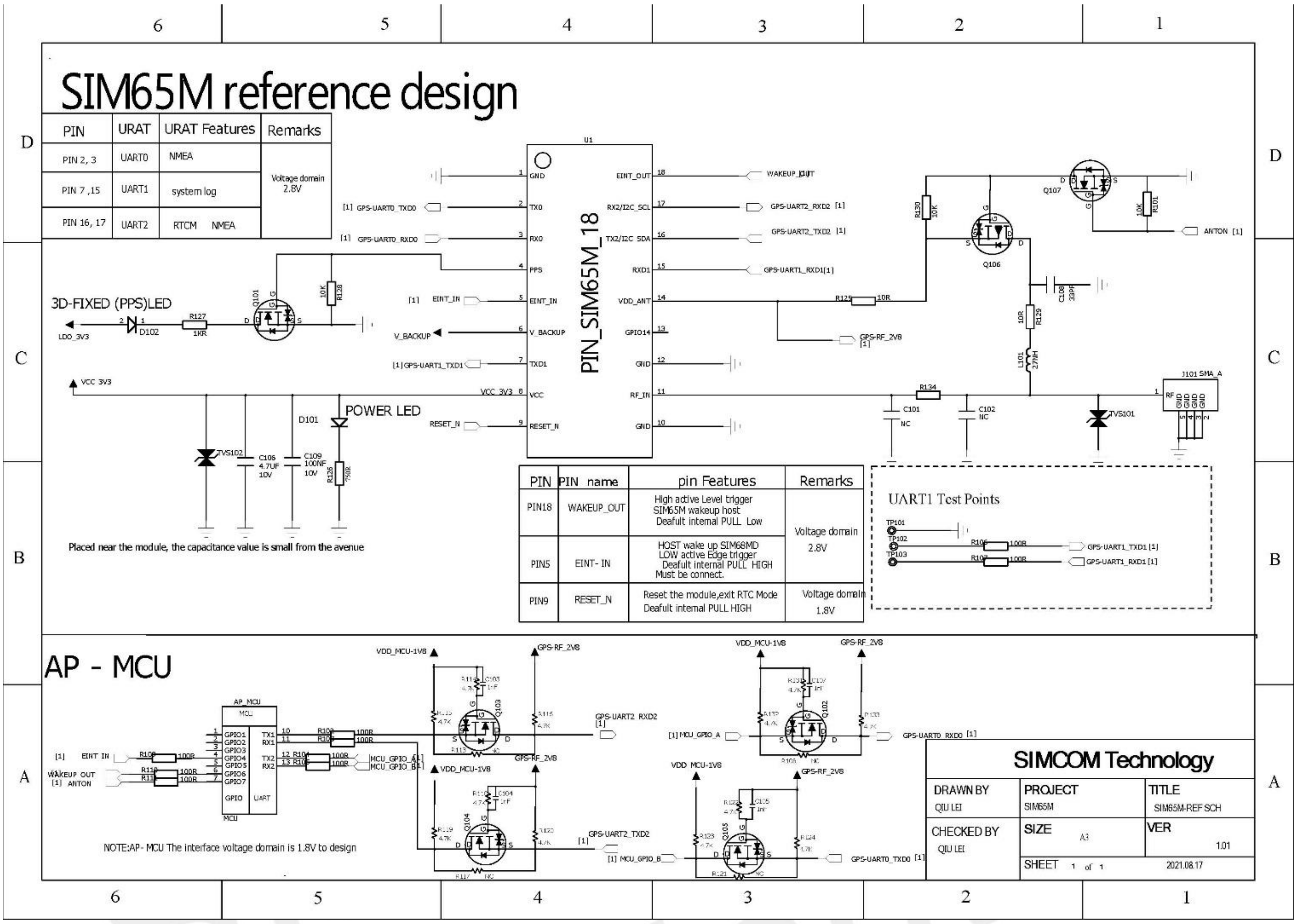

Figure 10: SIM65M antenna detection reference design

5. Appendix

5.1 Related Documents

Table 8: Related documents

SN	Document name	Remark
[1]	SIM65M_Hardware_Design	SIM65M Hardware Design Document
[2]	SIM68M_Hardware_Design	SIM68M Hardware Design Document

5.2 Terms and Abbreviation

Table 9: Terms and Abbreviations

Abbreviation	Description
GNSS	Global Navigation Satellite System
NMEA	National Marine Electronics Association
IO	Input/Output
RF	Radio Frequency
RTC	Real Time Clock
RTCM	Radio Technical Commission for Maritime Services
UART	Universal Asynchronous Receiver & Transmitter
NC	Not connect
TTFF	Time To First Fix